Homework 1

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Question 1: A Simple Delegated Monitor Model

Consider a firm who needs \$100,000 to initiate a project. The payout of this project is \$200,000 with probability 90%, \$100,000 with probability 5%, and \$0 with probability 5% at the end of two years. There are 1,000 small investors each with \$100 to lend. If the project is initiated, only the firm manager knows the result of the project, while the lenders cannot see the results. However, as long as one of the lenders pays a monitor fee of \$200, the results will be known to everyone.

Part (a)

Assume each investor knows that there are many other small investors with \$100 to lend, but each investor lives on an isolated island and cannot coordinate with other investors; he or she can only communicate with the firm. Suppose the project is the only investment opportunity for the small investors. Will the small investors lend directly to the firm? Will the small investor pay the monitor fee? Why or why not?

Step 1: Will investors pay the monitoring fee?

To determine whether any investor would pay the \$200 monitoring fee, we analyse the strategic decision using the following Nash Equilibrium payout matrix:

Your Decision	Others Monitor	Others Don't Monitor
Monitor	Your share − \$200	Your share − \$200
Don't Monitor	Your share − \$0	\$0 (firm claims failure)

If others are already monitoring, the results are publicly revealed and paying \$200 is redundant; the dominant strategy is not to monitor. If others are not monitoring, the firm manager can freely claim the project failed and pocket the proceeds. An investor's expected share is only $0.001 \times (\$200{,}000 \times 0.9 + \$100{,}000 \times 0.05 + \$0 \times 0.05) = \$185$, which is less than the \$200 monitoring cost. Even if one investor were willing to pay, the free-rider problem ensures no single investor voluntarily bears the full cost. The unique Nash Equilibrium is therefore that **nobody monitors**.

Step 2: Will investors lend to the firm?

Given that no monitoring will occur, the firm manager always has the incentive to claim the project failed and retain all proceeds. A rational investor anticipates this and expects a return of \$0 on any \$100 lent. Since keeping the \$100 yields \$100 with certainty while lending yields \$0, **no small investor will lend to the firm**. This outcome is a classic market failure driven by the combination of information asymmetry and coordination failure: each investor hopes another will pay the monitoring fee, nobody does, and the project goes unfunded.

Part (b)

Will the project be funded if the 1,000 small investors can communicate with each other? If so, explain how they can do it and what is the expected rate of return on the investment.

Yes, the project will be funded once communication is possible, because investors can solve both the free-rider problem and the monitoring cost problem through coordination.

Option 1 — Shared Monitoring Cost. All 1,000 investors agree to contribute \$0.20 each toward monitoring, raising the required \$200 in total. Each investor's total outlay becomes \$100.20. With monitoring guaranteed, the firm manager can no longer misreport results, and the investment is viable.

Option 2 — Delegated Monitor (Financial Intermediary). Investors pool their funds and appoint a single intermediary (a bank) that collects \$100,000 and pays the \$200 monitoring fee on their behalf. The intermediary acts as a single lender with full monitoring capability, illustrating the fundamental role of banks as delegated monitors.

Expected Rate of Return Calculation. The gross expected payout of the project is:

$$\mathbb{E}[\text{Payout}] = \$200{,}000 \times 0.9 + \$100{,}000 \times 0.05 + \$0 \times 0.05 = \$185{,}000.$$

The total cost of the investment (principal plus monitoring) is $\$100{,}000 + \$200 = \$100{,}200$. The net profit is therefore $\$185{,}000 - \$100{,}200 = \$84{,}800$, giving a return over two years of:

$$\frac{\$84{,}800}{\$100{,}200} \approx 84.63\%.$$

The annualised return is approximately $(1.8463)^{0.5} - 1 \approx 36.0\%$ per year. Each investor who contributed \$100.20 can expect to receive approximately \$185 after two years, a dramatically better outcome than the \$0 return available without coordination.

Question 2: Diversification of Risk through Financial Intermediation

In an economy there are many identically distributed and independent projects with a one-year investment horizon. The net rate of return on each project follows a normal distribution with mean $\mu = 6\%$ and standard deviation $\sigma = 15\%$. Each project requires an initial investment of \$1,000. Formally, $r \sim N(\mu, \sigma^2)$ with $\mu = 6\%$ and $\sigma = 15\%$.

Part (a)

Suppose a bank invests in only one project. What is the probability of losing more than half of the initial investment (\$500)? What is the probability of losing more than 10% of the initial investment (\$100)?

A loss of more than \$500 corresponds to a return below -50%. The standardised z-score is:

$$z = \frac{-0.50 - 0.06}{0.15} = \frac{-0.56}{0.15} \approx -3.733,$$

so $P(r < -50\%) = \Phi(-3.733) \approx 0.0094\%$.

A loss of more than \$100 corresponds to a return below -10%. The standardised z-score is:

$$z = \frac{-0.10 - 0.06}{0.15} = \frac{-0.16}{0.15} \approx -1.067,$$

so $P(r < -10\%) = \Phi(-1.067) \approx 14.31\%$. There is therefore a non-trivial 14.3% chance that a single-project bank loses more than 10% of its investment.

Part (b)

Suppose the bank invests in 15 such projects. What is the mean and standard deviation of the rate of return? What is the probability of losing more than 10% of the total initial investment (\$1,500)?

By the Central Limit Theorem, the average return across 15 i.i.d. projects is:

$$\bar{r}_{15} \sim N\left(\mu,\ \frac{\sigma^2}{N}\right), \quad \mu = 6\%, \quad \frac{\sigma}{\sqrt{15}} = \frac{15\%}{\sqrt{15}} \approx 3.873\%.$$

A loss of more than \$1,500 on the \$15,000 total investment requires the average return to fall below -10%:

$$z = \frac{-0.10 - 0.06}{0.03873} \approx -4.131,$$

so $P(\bar{r}_{15} < -10\%) = \Phi(-4.131) \approx 0.00180\%$. Diversifying across 15 projects reduces the probability of a 10% loss from 14.3% to just 0.0018%—a dramatic reduction in risk illustrating the power of portfolio diversification.

Part (c)

What is the mean and standard deviation of the rate of return if the bank invests in N projects? What happens as $N \to \infty$?

For a portfolio of N independent, identically distributed projects, the average return satisfies:

$$\bar{r}_N \sim N\left(\mu,\ \frac{\sigma^2}{N}\right).$$

The mean remains constant at $\mu = 6\%$ regardless of N, while the standard deviation equals $\sigma/\sqrt{N} = 15\%/\sqrt{N}$, which decreases monotonically toward zero as N grows. As $N \to \infty$, the Law of Large Numbers guarantees that the portfolio return converges almost surely to the mean:

$$\bar{r}_N \xrightarrow{\text{a.s.}} \mu = 6\%.$$

This is the core insight of financial intermediation: a bank holding infinitely many uncorrelated projects faces zero variance risk, and the portfolio return becomes a certainty of 6%.

Part (d)

Suppose the bank's assets consist of infinitely many such projects ($N \to \infty$), and it offers depositors a one-year certificate of deposit (CD) with a fixed gross return $1+r$. What is the maximum interest rate a fully competitive bank (zero profit) can offer? Ignore operational costs.

When $N \to \infty$, the bank earns exactly $\mu = 6\%$ on its asset portfolio with no uncertainty. A fully competitive bank earns zero profit by passing the entire return to depositors:

$$\text{Maximum CD rate} = \mu = 6\%.$$

The bank has transformed risky individual loans (each with a 15% standard deviation) into a riskless liability—a fixed-return CD—by pooling and diversifying. This perfectly illustrates how banks create safe secondary securities (deposits) from risky primary assets (loans), which is a fundamental function of financial intermediation.

Question 3: Bank Run

Consider a simplified version of the Diamond–Dybvig (1983) model. A bank collects deposits of 100 from many identical depositors at date 0. At date 1, a fraction of 20% of depositors are impatient and must consume early; the remaining 80% are patient and can wait until date 2. The bank invests all funds in a long-term project that yields 1.4 per unit if held to date 2, or 0.8 per unit if liquidated early at date 1. The demand deposit contract promises 1.0 per depositor who withdraws at date 1 and 1.2 per depositor who waits until date 2. The setup is summarised below.

Parameter	**Value**
Total deposits (date 0)	100
Impatient depositors (withdraw date 1)	20% $\Rightarrow$ 20 depositors
Patient depositors (withdraw date 2)	80% $\Rightarrow$ 80 depositors
Long-term yield (held to date 2)	1.4 per unit
Early liquidation yield (date 1)	0.8 per unit
Deposit promise (date 1)	1.0 per depositor
Deposit promise (date 2)	1.2 per depositor

Part (a): Solvency under Normal Withdrawal

Suppose only the 20 impatient depositors withdraw at date 1. Is the bank solvent?

To meet the 20 units owed at date 1, the bank must liquidate $20/0.8 = 25$ units of its investment early, leaving 75 units intact. At date 2, those 75 units grow to $75 \times 1.4 = 105$ units, which comfortably covers the $80 \times 1.2 = 96$ units owed to patient depositors:

$$\text{Date-2 assets} = 105 \;\geq\; 96 = \text{Date-2 obligations.}$$

The bank is fully solvent and retains a surplus of 9 units. This demonstrates that a healthy bank faces only a *liquidity* mismatch between short-term liabilities and long-term assets, not an insolvency problem.

Part (b): Bank Run Equilibrium

(i) What happens if patient depositors panic and withdraw at date 1?

If patient depositors fear that others will run, their dominant strategy becomes withdrawing early as well. The bank now faces all 100 depositors demanding 1.0 at date 1, but its assets yield only 0.8 per unit when liquidated early. Total assets available amount to $100 \times 0.8 = 80$ units against obligations of $100 \times 1.0 = 100$ units, leaving a shortfall of 20 units. Under a sequential service constraint (first-come, first-served), the first 80 depositors in line each receive 1.0, while the last 20 depositors receive nothing and the bank fails:

$$\text{Assets} = 80 < 100 = \text{Obligations} \;\Rightarrow\; \text{Bank fails.}$$

(ii) Why is this a self-fulfilling equilibrium?

This outcome is a self-fulfilling prophecy because the belief itself causes the outcome it predicts. If a patient depositor believes others will run, it is rational to withdraw early and receive 1.0 rather than risk receiving 0 later. Because enough depositors act on this belief, the bank truly fails, which retroactively justifies the original fear. The model therefore exhibits two Nash Equilibria: a *good equilibrium* in which patient depositors wait, the bank remains solvent, and everyone is paid according to the contract; and a *bad equilibrium* (bank run) in which everyone withdraws at date 1

and the bank collapses. Neither equilibrium depends on the bank's fundamental health—the bad equilibrium is driven purely by coordination failure and panic, which is Diamond and Dybvig's central insight.

Part (c): Policy Interventions

(i) How does deposit insurance change the equilibrium?

Deposit insurance eliminates the bad equilibrium entirely by removing the incentive to run. With government-guaranteed deposits, a patient depositor knows they will receive 1.2 at date 2 regardless of what others do. Withdrawing early yields only 1.0, so waiting is always the dominant strategy and no rational depositor will run. The key mechanism is that deposit insurance breaks the strategic complementarity: your optimal action no longer depends on what you fear others will do. The cost is moral hazard—banks may take excessive risks knowing deposits are guaranteed—which is why deposit insurance is typically paired with prudential regulation and capital requirements.

(ii) Would a lender of last resort (LOLR) eliminate the bad equilibrium?

A central bank acting as lender of last resort can also eliminate the bad equilibrium, but only under specific conditions. It must lend freely at a penalty rate (to ensure only solvent-but-illiquid banks borrow), against good collateral (to limit central bank risk), and the bank must be illiquid rather than insolvent. Crucially, the mere credible announcement of LOLR support can prevent a run from starting entirely. In this model the bank is solvent (as shown in Part a), so a LOLR pledge that the central bank will provide liquidity if needed makes it irrational for patient depositors to run—they know the bank can always honour 1.2 at date 2. If the bank were truly insolvent, however, LOLR support would merely delay failure and create moral hazard without resolving the underlying problem.